VANISHING OF HIGHER DIRECT IMAGES ABOVE THE FIBRE DIMENSION

THE STACKS PROJECT AUTHORS

Editorial note. No omissions; the chain is fully cited and each reduction argued.

Allowed standard prerequisites (editorial).

- Noetherian vanishing above fibre dimension. Cohomology of Schemes, Lemma coherent-lemma-higher-direct-images-zero-above-dimension-fibre

Source reference. Original lemma 0EX3: sources/stacks-chow/tags.txt, source lines 18643–18643. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02FZ: sources/stacks-chow/tags.txt, source lines 3210–3210. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 09ZR: sources/stacks-chow/tags.txt, source lines 12609–12609. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 05M5: sources/stacks-chow/tags.txt, source lines 7259–7259. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XC: sources/stacks-chow/tags.txt, source lines 2513–2513. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01F6: sources/stacks-chow/tags.txt, source lines 1859–1859. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XK: sources/stacks-chow/tags.txt, source lines 2521–2521. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01XJ: sources/stacks-chow/tags.txt, source lines 2520–2520. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 081F: sources/stacks-chow/tags.txt, source lines 10215–10215. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 089W: sources/stacks-chow/tags.txt, source lines 10511–10511. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 01PK: sources/stacks-chow/tags.txt, source lines 2233–2233. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01FF: sources/stacks-chow/tags.txt, source lines 1868–1868. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02V7: sources/stacks-chow/tags.txt, source lines 3758–3758. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let $y \in Y$. Assume f is proper and $\dim(X_y) = d$. Then*

1. *for $\mathcal{F} \in QCoh(\mathcal{O}_X)$ we have $(R^i f_* \mathcal{F})_y = 0$ for all $i > d$,*
2. *there is an affine open neighbourhood $V \subset Y$ of y such that $f^{-1}(V) \to V$ and d satisfy the assumptions and conclusions of Lemma 0EX3.*

Proof. By Morphisms, Lemma 02FZ and the fact that f is closed, we can find an affine open neighbourhood V of y such that the fibres over points of V all have dimension $\leq d$. Thus we may assume $X \to Y$ is a proper morphism all of whose fibres have dimension $\leq d$ with Y affine. We will show that (2) holds, which will immediately imply (1) for all $y \in Y$.

By Lemma 09ZR we can write $X = \lim X_i$ as a cofiltered limit with $X_i \to Y$ proper and of finite presentation and such that both $X \to X_i$ and transition morphisms are closed immersions. For some i we have that $X_i \to Y$ has fibres of dimension $\leq d$, see Lemma 05M5. For a quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ we have $R^p f_* \mathcal{F} = R^p f_{i,*}(X \to X_i)_* \mathcal{F}$ by Cohomology of Schemes, Lemma 01XC and Leray (Cohomology, Lemma 01F6). Thus we may replace X by X_i and reduce to the case discussed in the next paragraph.

Assume Y is affine and $f : X \to Y$ is proper and of finite presentation and all fibres have dimension $\leq d$. It suffices to show that $H^p(X, \mathcal{F}) = 0$ for $p > d$. Namely, by Cohomology of Schemes, Lemma 01XK we have $H^p(X, \mathcal{F}) = H^0(Y, R^p f_* \mathcal{F})$. On the other hand, $R^p f_* \mathcal{F}$ is quasi-coherent on Y by Cohomology of Schemes, Lemma 01XJ, hence vanishing of global sections implies vanishing. Write $Y = \lim_{i \in I} Y_i$ as a cofiltered limit of affine schemes with Y_i the spectrum of a Noetherian ring (for example a finite type $\mathbf{Z}$-algebra). We can choose an element $0 \in I$ and a finite type morphism $X_0 \to Y_0$ such that $X \cong Y \times_{Y_0} X_0$, see Lemma 01ZM. After increasing 0 we may assume $X_0 \to Y_0$ is proper (Lemma 081F) and that the fibres of $X_0 \to Y_0$ have dimension $\leq d$ (Lemma 05M5). Since $X \to X_0$ is affine, we find that $H^p(X, \mathcal{F}) = H^p(X_0, (X \to X_0)_* \mathcal{F})$ by Cohomology of Schemes, Lemma 089W. This reduces us to the case discussed in the next paragraph.

Assume Y is affine Noetherian and $f : X \to Y$ is proper and all fibres have dimension $\leq d$. In this case we can write $\mathcal{F} = \operatorname{colim} \mathcal{F}_i$ as a filtered colimit of coherent $\mathcal{O}_X$-modules, see Properties, Lemma 01PK. Then $H^p(X, \mathcal{F}) = \operatorname{colim} H^p(X, \mathcal{F}_i)$ by Cohomology, Lemma 01FF. Thus we may assume $\mathcal{F}$ is coherent. In this case we see that $(R^p f_* \mathcal{F})_y = 0$ for all $y \in Y$ by Cohomology of Schemes, Lemma 02V7. Thus $R^p f_* \mathcal{F} = 0$ and therefore $H^p(X, \mathcal{F}) = 0$ (see above) and we win. □